

artificial intelligence is the study of intelligent agents that perceive their environment and take actions to maximize the chances of achieving their goals. It is a branch of computer science focused on developing systems capable of performing tasks that typically requires human intelligence. for example reasoning , learning or problem solving.

Three main types of AI:

1. Narrow Ai: These type of AI are designed and trained to perform specific limited task. They cannot operate outside their pre-defined programming.
2. General Ai: A theoretical form of ai that would possess human-level cognitive abilities, meaning it could understand, learn and perform any intellectual task that a human can.
3. Super Ai: These type of ai is just a future concept, which is not implemented yet, and it is considered that, it would surpass the human intelligence, and can think, process and perform task which are impossible for a human to do.

Problem Formulation is the crucial process of defining an AI problem clearly before applying a search algorithm to solve it. it involves deciding what actions and states to consider given a specific goal, ensuring the Ai agent can search efficiently.

5 Components of a problem:

1. initial State: The starting position of the Agent.
2. Actions: The set of possible moves that are available to the agent.
3. transition model: A description of what happens after performing a specific action, essentially returning the resulting state.
4. Goal test: A condition that checks whether the current state matches the desired goal state.
5. path cost: A function measuring the total cost of the sequence of actions to reach the goal.

=====Reflex  
Agent=====Planning Agent

Behavior ===== Acts Strictly based on the current perception, ignoring future consequences.  
=====anticipates future outcomes by performing sequence of actions before acting.

Mechanism===== Uses Simple condition-Action(If-then) Rules.  
=====Explores all state space to find an optimal path to the desired goal state.

Speed===== Generally Very fast, as it involves no searching and planning.  
===== Slower, as it requires computational time to search and evaluate future states.

Example:=====Vaccum Cleaner. If Dirty square found, clean it, else move forward.  
=====Google Map: It finds all possible paths and plans the shortest path to the destination.

Search problem in AI: A search problem in ai is the challenge of finding a sequence of actions that transitions an agent from an Initial state to a desired goal state.

6 Components of a search problem:

1. Initial state: it is the starting position of the Agent.
2. State Space: The Set of all possible states of the problem.

3. Actions: The set of all possible moves from an state that are available to the agent
4. Transition model: A description of what happens after performing an action, essentially returning to the resulting state.
5. Goal Test: A condition that checks whether the current state matches the desired goal state.
6. Path cost: A function that measures the total cost for a sequence of actions to reach a goal state.

//////////State-Space

Graph=====

Search Tree

Definition: A Physical Representation of all possible states and their transitions.===== A Logical Structure Built During the search process to explore possibilities.

Nodes: Each Node Represents a unique state in the environment.  
===== Nodes Represents Explored States. Duplicate states can appear on different branches.

Structure: Fixed, contains all potential configurations and often includes cycles/loops=====  
Dynamic, Generated as the algorithm runs, Branches Represent possible Sequences of Actions.

Alpha-beta Pruning is an optimization technique for the minimax algorithm that eliminates branches of the search tree that cannot possibly affect the final decision. Alpha tracks the best value for the Max player, and beta tracks the best value for the Min player.

Efficiency Improvement: By pruning the useless branches, it reduces the number of nodes the algorithm explores, This faster search allows the ai to evaluate the search tree to a much deeper level within the same computational time limit.

//////////Uninformed

Search=====

===== Informed Search.

Alternative Name: Blind

search=====

===== Heuristic Search or Intelligent Search

Working: Has no problem-specific Heuristic Information, it solely uses the problem  
Definition.== Uses a Heuristic Function to estimate the cost to the goal. Guiding the Search  
Directions.

Speed: Explore More nodes, Making it Slower for large state space.  
===== Explore fewer nodes by prioritizing promising paths,  
making it faster.

Example: Breadth First Search(BFS), Depth First Search(DFS) Uniform Cost Search(UCS)  
===== A\* Search, Greedy Best First Search.

Uncertainty occurs when ai system lacks complete or accurate information from it's  
environment. This makes it impossible to predict the outcome of an action with absolute  
certainty.

It is primarily happens due to incomplete information, noisy sensor data, dynamic environments  
and random events. To Function under these conditions, ai system uses probabilistic reasoning  
to calculate and make the most rational, safest decision possible.

Learning rate is a critical hyperparameter in an optimization algorithm that controls the step size the model takes when updating its weights to minimize the loss function.

Problems:

Too high: When the learning rate is too large, The algorithm may overshoot the global minima, causing the loss to oscillate wildly or even diverge entirely.

Too low: When the learning rate is too small, The training process becomes extremely slow, and the model is highly susceptible to getting stuck in suboptimal local minima.

Prevention Technique:

Learning rate schedules: Initially taking a large learning rate, and then taking smaller learning rate as the algorithm runs.

Adaptive optimizers: Using optimization algorithms such as adam or adagrad to select the best learning rate for the loss function.

Grid search/Cross Validation: Testing a large combination of learning rates to find the most optimal one.

Genetic algorithms are search and optimization heuristics inspired by charles darwins theory of natural evolution. it reflects the process of natural selection, where the fittest individuals are selected for reproduction and to produce offspring of the next generations.

working process:

Initialization: Initialize a population of random individual.

Fitness evaluation.

Selection.

Crossover/Recombination.

Mutation.

Termination.

Example: Traveling salesperson problem.

Backtracking is an algorithmic technique for solving problems recursively by trying to build a solution incrementally, one piece at a time. It uses a depth-first search strategy, when the algorithm realizes that the current partial solution cannot possibly lead to a complete, valid solution, it backtracks to the previous state and try a different path.

Overfitting occurs when a model learns the training data too well, memorizing all the specific pattern. as a result, it performs exceptionally well in the training data, but fails to generalize, leading to poor performance on a new unseen test data.

Prevention techniques:

Cross Validation: Using techniques like k-fold cross validation, makes the model understand the actual pattern of the data.

Regularization: Adding penalty to the loss function, such as L1, L2 regularization.

Large training dataset: big amount of training data makes the model learn the actual pattern and relationship, instead of memorizing.

Early stopping: Stopping the model before it learns all the complex structure of the training dataset.

Evaluation metrics are crucial because they provide an objective, mathematical measurement of how accurately a model performs its task.

Quantifying performance: They tell us the difference between the models prediction and the actual ground truth.

Comparing models: Helps us to evaluate multiple models or algorithms according to their evaluation metrics and then choosing the best one.

Hyperparameter tuning: Various types of hyperparameters are evaluated, and thus choosing the best combination for getting the minimum loss becomes possible.

Artificial neural networks is a computing system inspired by biological neural networks, consisting of interconnected nodes that are arranged in layers that process information using weighted connections to solve non linear problem.

Convolutional neural network is a specialized type of deep neural networks that uses convolutional layers to automatically extract spatial hierarchies of features. it is primarily used in image processing, object detection and computer vision tasks.

RNN: A class of artificial neural networks designed to process sequential data by maintaining a hidden state (memory) of previous inputs, allowing information to persist across time steps.

Bias Variance Tradeof: The tension in model building between the error introduced by simplistic assumptions (Bias, which causes underfitting) and the error introduced by excessive sensitivity to small fluctuations in the training set (Variance, which causes overfitting).